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Two-phase fully-coupled smoothed particle hydrodynamics (SPH) model for unsaturated soils and its application to rainfall-induced slope collapse

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ABSTRACT

In this paper, a new fully-coupled Smoothed Particle Hydrodynamics (SPH) formulation for unsaturated soils is developed to study the influence of rainfall infiltration on slope stability. The single-layer two-phase formulation is investigated in SPH for the first time to simulate the response of unsaturated soils. The use of a single set of particles improves the computational efficiency and facilitates the implementation of infiltration boundary conditions. The Drucker–Prager strain-softening model, with the use of Bishop's effective stress, is adopted as the soil's constitutive model. New extensions of a first-order consistent wall boundary treatment are proposed for the coupled hydro-mechanical problem to enforce non-slip/free-slip conditions for the soil phase and water phase. A novel stress diffusion algorithm for general application is introduced to smooth out the numerical noise in the stress field under large deformation. The accuracy of the formulated SPH model is examined with available analytical solutions and experimental data. The proposed numerical scheme is finally applied to the simulation of rainfall-induced slope collapse of an unsaturated slope with two different bedrock geometries. Results demonstrate that the geometry of the bedrock is shown to play an important role in the failure initiation and propagation of the collapse. It is found that the proposed model allows the investigation of both triggering and post-failure mechanism, providing a smooth stress field even at large deformations.

1. Introduction

Water infiltration such as rainfall can be considered as one of the main triggering factors of the slope failure. The process of water infiltration leads to the rise of pore water pressure and the decrease of matric suction in unsaturated soils, which in turns decreases the effective stress. As a consequence, the soil strength is weakened, and the equilibrium may not able to be maintained, resulting in instability of slopes such as landslides. Rainfall-induced slope failures involve complex hydrological and geomechanical processes that depend on the geometries and the initial state of the slope, the hydro-mechanical properties of the soils and the infiltration parameters (Sorbino and Nicotera, 2013). In the last decades, substantial experimental efforts have been carried out to advance the understanding of the physics involved in rainfall-induced slope instabilities (e.g., Ochiai et al., 2004, Huang et al., 2008, Cui et al., 2014, Wang et al., 2020). It is recognised that a typical rainfall-induced landslide process can be divided into two main stages: onset of failure and post-failure stage (Cascini et al., 2010). Before the onset of failure, water infiltration modifies the stress state within the material, leading to a gradual development of one or more sliding surfaces. The failure initiates once a continuous sliding surface is formed. This is followed by the post-failure stage, which is characterized by the rapid accumulation of plastic strains and the sudden acceleration of the failed soil mass moving downslope.

Apart from experimental works, predictive (e.g., Guzzetti et al., 2008, Wu et al., 2015, Sasahara, 2017), statistical (e.g., Ibsen and Casagli, 2004, Chang and Chiang, 2009, Kristo et al., 2017), probabilistic (e.g., Zhang et al., 2010, Ering and Babu, 2016), and numerical methods (e.g., Cai and Ugai, 2004, Yoo and Jung, 2006, Davies et al., 2014, Leshchinsky et al., 2015) have been used to assess the stability of unsaturated slopes, and to provide insights into the failure mechanism.

In the early days of the development of numerical approaches, the seepage analysis and slope stability analysis were performed separately (e.g., Cai and Ugai, 2004, Yoo and Jung, 2006). The accuracy and computational efficiency of this uncoupled framework depended strongly on the selected time increments. The integrated analysis of hydraulic and mechanical response of unsaturated soil were implemented in later research works using a coupled framework (e.g., Yang

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et al., 2017, Senthilkumar et al., 2018). It was found that the coupled scheme was able to produce an accurate prediction of the wetting front and better assessment of slope stability when compared to uncoupled scheme (Qi and Vanapalli, 2015, Oh and Lu, 2015).

In routine practice, conventional mesh-based methods (e.g., finite-element method) are usually adopted to solve the coupled governing equations for the slope stability analysis especially if the failure triggering mechanism is of interest. Typical examples of the numerical programs for such types of analysis include Plaxis (Galavi, 2010), Geostudio (Krahn, 2012), and ICFEP (Potts et al., 2001). However, such methods may not be suitable to simulate the large deformation achieved in the post-failure stage since they tend to become unstable, and suffer from mesh distortions problems. Nevertheless, the post-failure analysis of a slope is of particular interest for disaster prevention and mitigation planning when assessing the potential consequences of landslides (e.g., run-out distance, landslide volumes and its potential impact on infrastructure). Therefore, there is a need to investigate novel numerical schemes that can capture the initiation of the slope failure as well as its post-failure response.

The emergence of meshless methods provides a promising solution for such analyses owing to their ability to incorporate the coupled multi-physics and capability of simulating large deformations. Examples of meshless methods used for geotechnical applications include: smoothed particle hydrodynamics (SPH), the material point method (MPM), and the particle finite element method (PFEM). Among them, SPH can be regarded as a truly meshless technique, whereas the other methods are still, to some extent, dependent of the background mesh. It is also worth mentioning the discrete element method (DEM), which is another popular numerical method in geomechanics to treat large deformation. However, compared to continuum-based methods such as SPH, the DEM simulation is computationally demanding, especially for real-scale problems involving millions of particles. SPH was originally invented to solve astrophysical problems by Gingold and Monaghan (1977), and Lucy (1977). Following the success of SPH application to fluid and solid mechanics (Libersky et al., 1993, Monaghan, 1994), more and more applications of SPH have been made in recent years for a vast range of engineering problems, such as fracture of solids (Benz and Asphaug, 1995), multi-phase flow (Colagrossi and Landrini, 2003), plastic flow (Zhou et al., 2007), and geophysical flows (Bui and Nguyen, 2021).

For the coupled hydro-mechanical problems, two types of SPH framework are available to describe the soil–water interactions. The first framework is the so-called two-phase double layer approach, which uses two layers of SPH particles to represent the soil phase and water phase respectively. The fluid particles and solid particles are overlapped, and move according to their own governing equations. The second framework is the two-phase single layer approach, in which the computational domain is discretized into a single set of particles carrying the information of both soil and water phases that moves according to the material velocity of the soil phase. The two-phase double layer formulation is able to reproduce the fluid regime both inside and outside of the porous material and can be extended to a non-laminar fluid flow. It has been successfully used in several applications, including simulations of liquefied soils (Huang et al., 2013), wave-porous structure interaction (Ren et al., 2014), seepage failure of dikes subjected to water impoundment and rainfall (Zhang et al., 2016), stability analysis of soil slope (Bui et al., 2011, Zhang et al., 2019), submerged granular column collapse (Wang et al., 2017) and flow through porous media (Bui and Nguyen, 2017, Peng et al., 2017). However, the use of the two-phase double layer formulation is computational demanding, thus hindering its application to large-scale problems. By contrast, the use of two-phase single layer approach (e.g., Pastor et al., 2009, Blanc and Pastor, 2013) is computationally efficient. With the single layer approach, the boundary conditions, such as infiltration/evaporation or hydraulic head boundaries, can be easily implemented and it is straightforward to account for coupled multi-physics processes (Pinyol et al., 2018).

Although numerous mesh-free formulations have been proposed to solve hydro-mechanical problems involving large deformations and free-surface flows, their applications to the study of rainfall-induced slope instability are less common. Zhang et al. (2016) made the first attempt to adopt a double layer approach within SPH for modelling rainfall-induced failure of an unsaturated dike. The rainfall event was reproduced by generating rain particles with controlled time interval and velocities. Though satisfactory results were achieved in comparison with experimental observations, the algorithms for treating boundary condition were still cumbersome, and the computational costs were demanding. Another difficulty associated with the use of a double layer approach is the treatment of unsaturated soils in the capillary region. The kernel interpolation was commonly used to estimate the degree of saturation once a water particle was found at the kernel support of a soil particle (e.g., Zhang et al., 2016, Bui and Nguyen, 2017). As a result, the dimensions of the capillary rise were linked with the radius of the kernel support (or the smoothing length). The reliability of this treatment remains unclear as the smoothing length is a numerical parameter controlling the accuracy of SPH interpolations. Wang et al. (2018) proposed a single layer two-phase coupled MPM formulation to model the rainfall-induced slope collapse. The infiltration process was simulated by assigning zero pore water pressure to the free-surface particles. However, since the particles with prescribed zero pressure were allowed to move outside of the boundary regions, the prescribed pressure conditions along the free surface could not be maintained throughout the simulation. Consequently, an additional surface boundary algorithm was required to identify the locations of free-surface boundary cells/nodes/particles at each time step, and to re-apply pressure conditions for those particles (e.g., Bandara et al., 2016, Ceccato et al., 2021).

So far, the majority of particle-based continuum approaches for coupled hydro-mechanical problems tend to focus on the prediction of kinematics, resulting in reasonable predictions of the sliding surface and evolution of surface morphology (Bui and Nguyen, 2021). However, numerical oscillations and noise in the stress field were an issue in the post-failure stage (Wang et al., 2018), which could possibly undermine the accuracy of the simulation and consequently reduce the predictive capability of the model, especially for the applications involving soil-structure interactions, in which accurate predictions of forces/stress are of particular importance.

This work develops a new SPH formulation for unsaturated soils to study the slope instability induced by rainfall infiltration. The single-layer approach is adopted as it is known to be well suited for solving multi-physics problems. It is the first time for the single-layer formulation applied in SPH to analyse the coupling of unsaturated seepage flow and soil deformation. The advantages of the proposed method compared to existing double-layer SPH formulations include the improved computational efficiency, the reasonable description of the capillary region, a simple implementation of hydraulic boundary conditions, and more importantly a noise-free stress treatment. A first-order consistent wall boundary treatment developed by Feng et al. (2021) is extended for the boundary conditions of the coupled soil–water problem, including non-slip/free-slip conditions for both soil and water phase. It is shown that the zero-pressure boundary conditions are implicitly satisfied using the proposed formulation without the need of a surface boundary algorithm, thus simplifying the model's implementation. A new stress diffusion term is investigated to smooth out the numerical noise in the stress field at large deformation. Compared to the formulation by Feng et al. (2021), which is limited to the gravity-dominated failure (e.g., granular flow), the new method is more general and thus is applicable for diverse scenarios involving the rainfall-induced landslides, seepage-induced dam instabilities, or any other coupled hydro-mechanical problems. The validity of the present numerical scheme is examined through the infiltration test and drainage test; both cases show good agreement with the analytical solutions and experimental data. Finally, the proposed model is applied to study the rainfall-induced slope failure. Results show that the failure process from initiation to post-failure is well captured by the proposed model.

The remaining part of this paper is organised as follows: In Section 2, the two-phase coupled hydro-mechanical formulation for saturated/unsaturated soils and the adopted mechanical/hydraulic constitutive models are presented. This is followed by the SPH discretisation of governing equations together with the numerical implementation of initial/boundary condition, development of a numerical diffusion term, and time stepping scheme, given in Section 3. The validity of the proposed methodology is assessed in Section 4 where several validation cases, including infiltration test and drainage test of soil columns, are presented. In Section 5, the applications of the present SPH scheme to study the rainfall-induced slope failure are investigated. Section 6 provides a discussion of the limitations of the present model with suggestions for future works. Finally, some conclusions are drawn in Section 7.

2. Governing equations

Unsaturated soils are generally constituted by three phases, including soil phase, s, water phase, w, and air phase, a, as shown in Fig. 1. The soil phase formulates the soil skeleton, while the water and air phase fill the voids between the soil particles. In many applications, the presence of the air phase can be considered in a simplified way by assuming that the air pressure is zero and the air density is negligible compared to those of the water and soil phases (Bandara et al., 2016, Wang et al., 2018, Lei et al., 2020, and Ceccato et al., 2021). Thus, the governing equations for air phase can be omitted and the three-phase mixture can be described using a two-phase model. Unsaturated conditions are taken into account by considering the evolution of the water content and the presence of capillary pressure.

The two-phase formulations for the unsaturated soil can be derived based on the mixture theory as described by Wang et al. (2018), Lei et al. (2020), Ceccato et al. (2021). Alternatively, the formulation can be derived based on Biot's formulation (Biot, 1941) and its extensions (Zienkiewicz et al., 1990) as in the works by Sheng et al. (2003), Khalili et al. (2008), Schrefler and Scotta (2001), Bandara et al. (2016). In this study, the mixture theory is adopted, the governing equations are obtained based on the mass balances of both water phase and soil phase, dynamic momentum balance of the mixture, the linear momentum equation, i.e., Darcy's law, of the water, and the hydraulic/mechanical constitutive relationships. The material coordinates are assumed to be attached to the soil skeleton and the motion of fluid flow is described with respect to the soil skeleton. The derivation of the governing equations is based on the following sign convention: (i) the compressive stress and strains are negative for the soil phase; (ii) pore water pressures in the water phase are assumed to be positive in compression; (iii) the flow discharge is considered to be positive for inflow.

2.1. Mass balance equations

The material derivative with respect to the soil phase takes the following generic form,

$$\frac{D^s(\cdot)}{Dt} = \frac{\partial(\cdot)}{\partial t} + \boldsymbol{v}_s \cdot \nabla(\cdot), \tag{1}$$

where superscript s denotes soil phase, and $\boldsymbol{v}_s$ is the soil phase velocity.

The mass balance equation of soil phase can be written as,

$$\frac{\partial n_s \rho_s}{\partial t} + \nabla\cdot(n_s \rho_s \boldsymbol{v}_s) = 0, \tag{2}$$

where ρ_s is intrinsic density of soil grains, n_s is the volumetric fraction of soil phase, and it can be related to porosity n by $n_s = 1 - n$.

Assuming the soil grain to be incompressible (hence ρ_s is a constant) and neglecting the spatial variation of porosity, Eq. (2) can be rewritten as,

$$\frac{D^s n}{Dt} = (1 - n)\nabla\cdot\boldsymbol{v}_s. \tag{3}$$

The mass balance equation of water phase is given by,

$$\frac{\partial n_w \rho_w}{\partial t} + \nabla\cdot(n_w \rho_w \boldsymbol{v}_w) = 0, \tag{4}$$

where ρ_w is the water density, n_w is the volumetric fraction of water phase, which can be related to porosity by the degree of saturation S using $n_w = nS$, and $\boldsymbol{v}_w$ is the water velocity.

Including Eq. (3) into Eq. (4), and rearranging the obtained equation gives (Ceccato et al. 2021),

$$n\frac{D^s(\rho_w S)}{Dt} = \nabla\cdot(\rho_w n S \boldsymbol{v}_{sw}) - S\rho_w \nabla\cdot\boldsymbol{v}_s. \tag{5}$$

If the spatial variation of water density is further considered to be negligible (e.g., Lei et al., 2020), taking into account the definitions of seepage discharge, i.e., $\boldsymbol{q}_w = nS\boldsymbol{v}_{ws}$, Eq. (5) can be rewritten as,

$$-n\frac{\partial S}{\partial p_c}\frac{D^s p_w}{Dt} + nS\frac{1}{\rho_w}\frac{D^s \rho_w}{Dt} + S\nabla\cdot\boldsymbol{v}_s = -\nabla\cdot\boldsymbol{q}_w, \tag{6}$$

where p_c is the suction pressure, defined as $p_c = p_a$-p_w, which is taken as -p_w since the air pressure p_a is assumed to be zero in the analysis. Note that equation (6) has three storage terms on the left-hand side, related to retention curve, water compressibility, and soil skeleton compressibility, respectively, from left to right.

In this study, density rather than pressure is taken as the state variable. This selection is consistent with the weakly compressible assumption of the water, in which water density shows a slight variation due to the movement of water (Monaghan, 1994). The equation of state

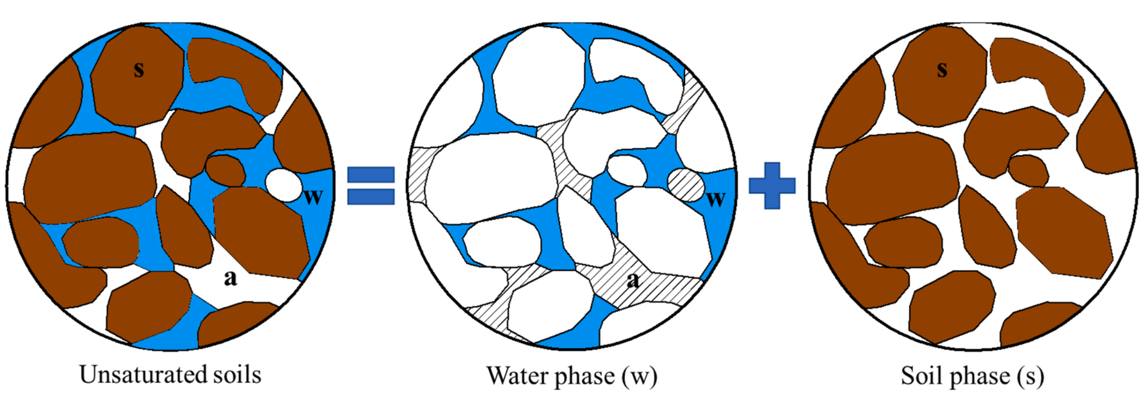

Fig. 1. Representation of unsaturated soils (the hatched regions in the water phase represent the air phase).

is adopted to link density with pressure. The following relationship is established for the first term on the left-hand side of Eq. (6), given by,

$$\frac{D^s p_w}{Dt} = \frac{K_w}{\rho_w}\frac{D^s \rho_w}{Dt}, \tag{7}$$

where K_w is the bulk modulus of water.

Substituting Eq. (7) into Eq. (6), yields the equation of liquid density derivative,

$$n\left(S - K_w\frac{\partial S}{\partial p_c}\right)\frac{D^s \rho_w}{Dt} = -\rho_w(\nabla\cdot\boldsymbol{q}_w + S\nabla\cdot\boldsymbol{v}_s). \tag{8}$$

The derivative of the saturation degree with respect to the suction pressure is governed by the soil–water characteristic curve (SWCC) given later in Section 2.4.

It is noted that the final formulations of the mass balance equation for water phase, i.e., Eq. (8) are the similar to that presented by Bandara et al., (2016) if the Biot's coefficient is set to 1, and similar to that based on the mixture theory as in Lei et al., (2020). Differences from other established formulations, such as those by Gatmiri et al. (1998), Sheng et al. (2003), Khalili et al. (2008), Tsiampousi et al. (2017a) lie in the expressions of the storage term related to soil skeleton compressibility or retention curve. In addition, the formulation of mass balance equation is consistent with the adopted SWCC model where the degree of saturation is solely related to the suction pressure, as shown in Tsiampousi et al. (2017a,b). However, Eq. (8) should be modified to implement more complex SWCC models, such as those that consider the effect of volume change on the degree of saturation (See Tsiampousi et al., 2017a).

2.2. Momentum balance equations

The momentum conservation for the water phase can be written as,

$$\rho_w\boldsymbol{a}_w = -\nabla p_w - \boldsymbol{f}_d + \rho_w\boldsymbol{b}, \tag{9}$$

where $\boldsymbol{a}_w$ is the water acceleration, $\boldsymbol{f}_d$ is the drag force exerted on solid phase which describes the solid–fluid interaction, p_w is the pore water pressure, and $\boldsymbol{b}$ is the body force.

The flow within the porous material is assumed to be laminar and stationary. Thus, the drag force term is governed by the Darcy's law, taking the following form,

$$\boldsymbol{f}_d = \frac{n_w\gamma_w}{K}(\boldsymbol{v}_w - \boldsymbol{v}_s), \tag{10}$$

where K is the hydraulic conductivity at a given degree of saturation S and γ_w is the unit weight of the water.

The momentum conservation for the mixture is given by,

$$n_s\rho_s\boldsymbol{a}_s + n_w\rho_w\boldsymbol{a}_w = \nabla\cdot\boldsymbol{\sigma} + \rho_m\boldsymbol{b}, \tag{11}$$

where $\boldsymbol{a}_s$ is the soil acceleration, and $\rho_m = n_s\rho_s + n_w\rho_w$ is the density of the mixture and $\boldsymbol{\sigma}$ is the total stress tensor.

The above momentum equations are established based on the velocities of the soil phase and the velocities of water phase, in which the dynamic terms for both the water phase and soil phase are taken into account. In this study, to simplify the formulation and for the convenience of the wall boundary treatment, the Darcy's velocity $\boldsymbol{q}_w$ and soil velocity $\boldsymbol{v}_s$ are taken as primary unknown instead of water velocity $\boldsymbol{v}_w$.

Under the assumption that the relative acceleration of water with respect to the soil skeleton is neglected and rearranging Eq. (9), the momentum equations for water phase and the mixture are respectively given by,

$$\boldsymbol{q}_w = \frac{K}{\gamma_w}[-\nabla p_w + \rho_w(\boldsymbol{b} - \boldsymbol{a}_s)], \tag{12}$$

and,

$$\rho_m\boldsymbol{a}_s = \nabla\cdot\boldsymbol{\sigma} + \rho_m\boldsymbol{b}. \tag{13}$$

The variation of hydraulic conductivity with respect to the degree of saturation (or suction pressure) is governed by the hydraulic conductivity curve (HCC) given later in Section 2.4.

2.3. Mechanical constitutive equations

The total stress tensor $\boldsymbol{\sigma}$ is calculated using the concept of Bishop's effective stress for unsaturated soils, taking the form,

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' - \chi p_w\boldsymbol{I}, \tag{14}$$

in which $\boldsymbol{\sigma}'$ is the effective stress tensor, χ is an effective stress parameter that varies from 0 to 1 to account for the dry to fully saturated conditions; for convenience, the value of χ can be taken equal to the degree of saturation S; $\boldsymbol{I}$ is the unit vector.

In this study, the two stress variables $\boldsymbol{\sigma}'$ and p_w are evaluated separately. The effective stress is computed from strain increment based on the elasto-plastic analysis, while the pore pressure is computed from the density variation assuming the pore fluid is weakly compressible. The stress increment is calculated by,

$$\frac{D\boldsymbol{\sigma}'}{Dt} = \boldsymbol{D}^{ep}\frac{D\boldsymbol{\varepsilon}}{Dt}, \tag{15}$$

where $\boldsymbol{D}^{ep}$ is the elasto-plastic stiffness tensor, which depends on the constitutive model adopted, $\boldsymbol{\varepsilon}$ is the strain tensor. The elastoplastic constitutive model is used in present analysis with Drucker-Prager yield criterion, in which the yield function and plastic potential function are given by,

$$f = \alpha_\phi I_1 + \sqrt{J_2} - k_c, \tag{16a}$$

$$g = \alpha_\psi I_1 + \sqrt{J_2}, \tag{16b}$$

where I_1 and J_2 are respectively the first principal stress invariant and the second deviatoric stress; α_ϕ and k_c are Drucker-Prager constants, which can be related to Coulomb's material constants c (cohesion) and ϕ (internal friction) according to,

$$\alpha_\phi = \frac{\tan\phi}{\sqrt{9 + 12\tan^2\phi}}, \tag{17a}$$

$$k_c = \frac{3c}{\sqrt{9 + 12\tan^2\phi}}, \tag{17b}$$

and, α_ψ is a dilatancy factor. Its value is related to the dilation angle ψ in a similar manner to that between α_ϕ and friction angle φ (Bui et al., 2014, Nguyen et al., 2017).

To account for the decrease of strength due to the large deformations developed at the post-failure stage, the cohesion is assumed to decrease exponentially with the plastic strain according to the non-linear cohesion-softening relationship (Bui et al., 2021),

$$c(\kappa) = c_r + (c_p - c_r)e^{-\eta_c\kappa}, \tag{18}$$

where c_p and c_r denote the peak and residual cohesion, respectively. η_c is the softening coefficient that controls the rate of strength degradation and κ is the internal variable, which is related to the equivalent plastic shear strain tensor according to,

$$d\kappa = \sqrt{\frac{2}{3}d\boldsymbol{e}^p : d\boldsymbol{e}^p}, \tag{19}$$

where $d\boldsymbol{e}^p$ is the deviatoric plastic strain increment.

In the present work the strain-softening response is modelled for the cohesion strength only. Yet, the same strain-softening response can be extended to the angle of shearing resistance using the formulation presented in Bui et al. (2021). It is also worth mentioning that meshless approaches, such as SPH, have a distinct attraction in comparison to traditional mesh-based approaches with their ability to handle large deformations efficiently. This advantage provides benefits in predicting phenomena such as strain-softening since the strain-softening behaviour of soils is linked to the post-failure that naturally involves large deformations. The use of meshless methods therefore enables the strain-softening at post-failure to be easily simulated without extra computational effort (such as coupling with another methodology or a remeshing technique). A typical example is the simulation of retrogressive slope failure of clayey slopes, which is challenging for mesh-based methods but can be easily achieved in meshless methods (e.g., Wang et al., 2016, Bui et al., 2021).

Compressibility is introduced such that the pressure can be linked to density. A compressible equation of state (EoS) presented by Morris et al. (1997) is adopted to calculate pressure as,

$$p_w = c_0^2(\rho_{\mathrm{w}} - \rho_{\mathrm{w0}}), \tag{20}$$

where ρ_{w0} is the reference density of the water and c_0 is the numerical speed of sound. The bulk modulus of water K_{w} for Morris EoS is given by,

$$K_w = c_0^2\rho_{\mathrm{w0}}. \tag{21}$$

2.4. Hydraulic constitutive equations

In addition to the mechanical constitutive model, the constitutive relationship between the degree of saturation and the suction pressure is needed to complete the governing equations. This is given by the soil–water characteristic curve (SWCC) for a particular soil. Two alternative models, i.e., linear model and Van Genuchten model, are considered in present analysis, whose formulations are given by Eqs. (22a) and (22b), respectively,

$$S = \begin{cases} S_{\max} & p_c \leqslant 0 \\ S_{\max} - a_v p_c & 0 < p_c < p_{cs} \\ S_{\min} & p_c \geqslant p_{cs} \end{cases}, \tag{22a}$$

$$S = S_{\min} + (S_{\max} - S_{\min})\left[1 + \left(\frac{p_{\mathrm{c}}}{p_{ref}}\right)^{\frac{1}{1-\lambda}}\right]^{-\lambda}, \tag{22b}$$

where a_{v}, λ, and p_{ref} are fitting parameters, $S_{\min}$ is the residual saturation degree, $S_{\max}$ is the maximum saturation degree, and p_{cs} is the threshold suction value to reach the residual saturation degree, whose value is equal to $p_{cs} = (S_{\max} - S_{\min})/\ a_{\mathrm{v}}$.

To take into account the effect of the degree of saturation on the hydraulic conductivity of the soil, the permeability of the unsaturated soil is expressed using the relative hydraulic conductivity K_{rel}, which is defined as the ratio of the hydraulic conductivity at a given degree of saturation to the saturated hydraulic conductivity, $K_{rel} = K/K_{sat}$. The hydraulic conductivity curve (HCC) for linear relation model and Van Genuchten model are given by Eqs (23a) and (23b), respectively,

$$K_{rel} = \begin{cases} 1 & p_c \geqslant 0 \\ (K_{rel}^r)^{\frac{p_c}{p_{cr}}} & 0 < p_c < p_{cr} \\ K_{rel}^r & p_c \geqslant p_{cr} \end{cases}, \tag{23a}$$

$$K_{rel} = \sqrt{S}\left[1 - \left(1 - S^{1/\lambda}\right)^{\lambda}\right]^2, \tag{23b}$$

where p_{cr} is the value of suction at which the relative permeability is reduced to the residual permeability K_{rel}^r. The value of K_{rel}^r can be taken as 10^{-4} as suggested by Galavi (2010).

3. Numerical implementations: Two-phase single layer formulation

3.1. SPH discretization

The applied the governing equations are summarised as follows,

$$\frac{D^s n}{Dt} = (1-n)\nabla\cdot\boldsymbol{v}_{\mathrm{s}}, \tag{24a}$$

$$n\left(S - K_{\mathrm{w}}\frac{\partial S}{\partial p_c}\right)\frac{D^s\rho_{\mathrm{w}}}{Dt} = -\rho_{\mathrm{w}}(\nabla\cdot\boldsymbol{q}_{\mathrm{w}} + S\nabla\cdot\boldsymbol{v}_{\mathrm{s}}), \tag{24b}$$

$$\boldsymbol{q}_{\mathrm{w}} = \frac{K}{\gamma_{\mathrm{w}}}[-\nabla p_{\mathrm{w}} + \rho_{\mathrm{w}}(\boldsymbol{b} - \boldsymbol{a}_{\mathrm{s}})], \tag{24c}$$

$$\rho_{\mathrm{m}}\boldsymbol{a}_{\mathrm{s}} = \nabla\cdot\boldsymbol{\sigma} + \rho_{\mathrm{m}}\boldsymbol{b}. \tag{24d}$$

In SPH, the approximation of gradient of a scalar field f and divergence of a vector field $\boldsymbol{f}$ can be defined as (Mayrhofer et al., 2013),

$$\langle\nabla f\rangle_i = \sum_{j=1}^{N}\frac{m_j}{\rho_j}(f_j + f_i)\nabla_i W_{ij}, \tag{25a}$$

$$\langle\nabla\cdot\boldsymbol{f}\rangle_i = -\sum_{j=1}^{N}\frac{m_j}{\rho_j}(\boldsymbol{f}_i - \boldsymbol{f}_j)\cdot\nabla_i W_{ij}, \tag{25b}$$

where $\langle\ldots\rangle$ stands for the SPH interpolation, the subscript i and j denote the interpolating particle and its neighbours, respectively, $W_{ij} = W(\boldsymbol{x}_i\text{-}\boldsymbol{x}_j, h)$ and $f_j = f(\boldsymbol{x}_j)$ are respectively the value of the kernel function W and the function f at the particle j with the position $\boldsymbol{x}_j$, N is the number of particles distributed in the support domain, and m_j is the mass of the particle j. The reader is referred to Violeau (2012) and Violeau and Rogers (2016) for more detail on the derivation of the discrete SPH operators.

By using the SPH gradient operator and divergence operator expressed in Eqs. (25a) and (25b), and the Bishop's effective stress formulation expressed in Eq. (14), the discretised form of Eqs. (24a) - (24d) takes following form, where the indicial notation is applied for convenience to represent the equations, in which α and β stand for the Cartesian coordinates,

$$\left\langle\frac{\mathrm{d}n}{\mathrm{d}t}\right\rangle_i = (n_i - 1)\sum_{j=1}^{N}\frac{m_{mj}}{\rho_{mj}}\left(v_i^\alpha - v_j^\alpha\right)\frac{\partial W_{ij}}{\partial x_i^\alpha}, \tag{26a}$$

$$\left\langle\frac{d\rho_w}{dt}\right\rangle_i = \left(S - K_w\frac{\partial S}{\partial p_c}\right)_i^{-1}\frac{\rho_{wi}}{n_i}\left(\sum_{j=1}^{N}\frac{m_{mj}}{\rho_{mj}}\left(q_{wi}^\alpha - q_{wj}^\alpha\right)\frac{\partial W_{ij}}{\partial x_i^\alpha} + S_i\sum_{j=1}^{N}\frac{m_{mj}}{\rho_{mj}}\left(v_i^\alpha - v_j^\alpha\right)\frac{\partial W_{ij}}{\partial x_i^\alpha}\right), \tag{26b}$$

$$\langle q_w^\alpha\rangle_i = \frac{K_i}{g}\left(-\frac{1}{\rho_{wi}}\sum_{j=1}^{N}\frac{m_{mj}}{\rho_{mj}}(p_i + p_j)\frac{\partial W_{ij}}{\partial x_i^\alpha} + g_i^\alpha - a_{si}^\alpha\right), \tag{26c}$$

$$\left\langle\frac{\mathrm{d}v_{\mathrm{s}}^\alpha}{\mathrm{d}t}\right\rangle_i = \frac{1}{\rho_{mi}}\sum_{j=1}^{N}\frac{m_{\mathrm{m}j}}{\rho_{\mathrm{m}j}}\left(\sigma_i^{\prime\alpha\beta} + \sigma_j^{\prime\alpha\beta}\right)\frac{\partial W_{ij}}{\partial x_i^\beta} - \frac{1}{\rho_{mi}}\sum_{j=1}^{N}\frac{m_{\mathrm{m}j}}{\rho_{\mathrm{m}j}}(S_i p_i + S_j p_j)\frac{\partial W_{ij}}{\partial x_i^\alpha} + g_i^\alpha, \tag{26d}$$

and the position of the particles is updated according to the following equation,

$$\frac{\mathrm{d}x_i^\alpha}{\mathrm{d}t} = \langle v_{\mathrm{s}}^\alpha\rangle_i, \tag{26e}$$

where m_{m} represents the mass of the mixture.

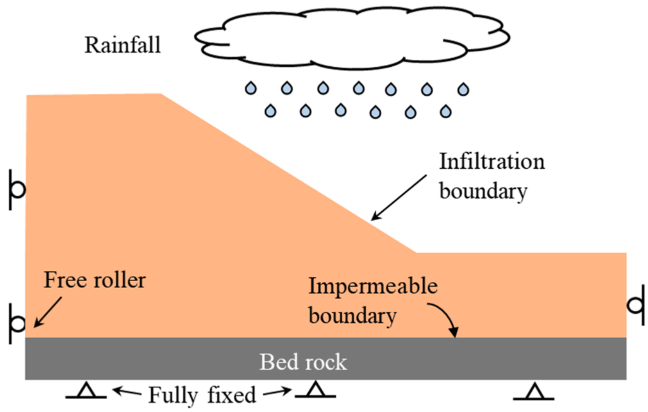

Fig. 2. Illustrative diagrams of boundary conditions for a slope subjected to rainfall-induced infiltration.

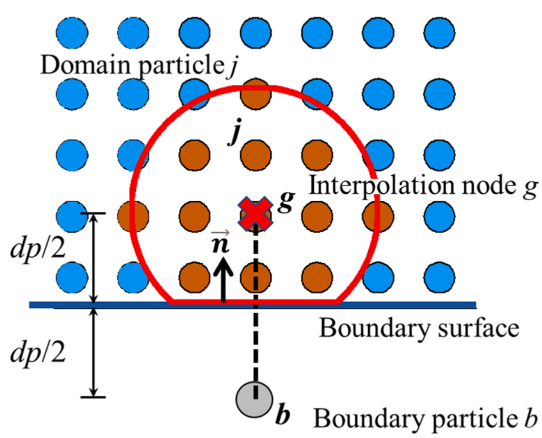

Fig. 3. Generation of the interpolation ghost nodes g.

Due to the zero-energy mode inherent to the SPH method, the solution is prone to unphysical oscillations and numerical instabilities when the numerical dissipative term is not included in the governing equations. This limitation can be overcome by introducing an artificial viscosity term into the momentum equation. This term was firstly proposed by Monaghan (1992) and has been found to stabilise the numerical algorithm. The artificial viscous term is introduced into the momentum equation as shown below,

$$\left\langle\frac{\mathrm{d}v^\alpha}{\mathrm{d}t}\right\rangle_i = \sum_{j=1}^{N}\frac{m_{\mathrm{m}j}}{\rho_{mi}\rho_{\mathrm{m}j}}\left(\sigma_i^{\prime\alpha\beta}+\sigma_j^{\prime\alpha\beta}+\Pi_{ij}\delta^{\alpha\beta}\right)\frac{\partial W_{ij}}{\partial x_i^\beta} - \sum_{j=1}^{N}\frac{m_{\mathrm{m}j}}{\rho_{mi}\rho_{\mathrm{m}j}}\left(S_ip_i + S_jp_j\right)\frac{\partial W_{ij}}{\partial x_i^\alpha} + g_i^\alpha, \tag{27}$$

where,

$$\Pi_{ij} = \begin{cases} \dfrac{\alpha_\Pi \bar{c}_{ij}\mu_{ij}}{\bar{\rho}_{ij}} & v_{ij}^\alpha x_{ij}^\alpha < 0 \\ 0 & v_{ij}^\alpha x_{ij}^\alpha > 0 \end{cases}, \tag{28a}$$

$$\mu_{ij} = \frac{h v_{ij}^\alpha x_{ij}^\alpha}{x_{ij}^\alpha x_{ij}^\alpha + \eta^2}, \tag{28b}$$

$$\bar{c}_{ij} = 0.5(c_i + c_j), \tag{28c}$$

$$\bar{\rho}_{ij} = 0.5(\rho_i + \rho_j), \tag{28d}$$

α_Π is the viscous constant controlling the magnitude of dissipation and normally takes values of 0.1 (Bui and Nguyen, 2021), $\eta = 0.1\ h$ is the numerical parameter introduced to avoid numerical singularity and c is the speed of sound. For soil materials it is computed by,

$$c_i = \sqrt{E_i/\rho_i}, \tag{29}$$

where E is Young's modulus of the soil.

3.2. Boundary conditions

Specifying conditions on the boundaries of a problem is an essential component of the numerical analysis to constrain the solution. Fig. 2 presents the typical boundary conditions used to simulate the rainfall infiltration in a slope; these include infiltration and impermeable boundaries for the water phase, and free-roller/fully-fixed boundaries for the soil phase. In general, the applied boundary conditions can be divided into two types, i.e., solid boundary conditions and free-surface boundary conditions.

3.2.1 Solid boundary condition

A well-recognised issue of SPH is the truncation of kernel interpolation support of the particles near the boundary (Vacondio et al., 2020). The truncated interpolation may lead to inaccurate results and unphysical effects. These can be addressed by introducing boundary particles to complete the kernel support of the domain particles near the boundary. Herein, a particle-based boundary methodology proposed by Feng et al. (2021) for the simulation of granular flow is extended for the coupled soil–water problem to impose the free-slip/non-slip conditions on the solid boundaries.

In this treatment, fixed dummy boundary particles, b, are created near the boundaries to characterize the physical boundary of the simulated domain, and for each boundary particles, a ghost node, g, is mirrored into the domain perpendicular to the boundary surface to interpolate field variables from surrounding domain particles, j, as shown in Fig. 3. More details of the implementation can be found in Feng et al. (2021).

The value and gradient of a field property f (i.e., fluid density and stress) of the ghost nodes are computed using the first order consistent SPH interpolation proposed by Liu and Liu (2006) according to,

$$\boldsymbol{A}_g \cdot \begin{bmatrix} f_g \\ \partial_x f_g \\ \partial_y f_g \\ \partial_z f_g \end{bmatrix} = \begin{bmatrix} \sum_j f_j W_{gj} V_j \\ \sum_j f_j \partial_x W_{gj} V_j \\ \sum_j f_j \partial_y W_{gj} V_j \\ \sum_j f_j \partial_z W_{gj} V_j \end{bmatrix}, \tag{30}$$

where the renormalisation matrix $\boldsymbol{A}_g$ is,

$$\boldsymbol{A}_g = \begin{bmatrix} \sum_j W_{gj}V_j & \sum_j (x_j - x_g) W_{gj}V_j & \sum_j (y_j - y_g) W_{gj}V_j & \sum_j (z_j - z_g) W_{gj}V_j \\ \sum_j \partial_x W_{gj}V_j & \sum_j (x_j - x_g)\partial_x W_{gj}V_j & \sum_j (y_j - y_g)\partial_x W_{gj}V_j & \sum_j (z_j - z_g)\partial_x W_{gj}V_j \\ \sum_j \partial_y W_{gj}V_j & \sum_j (x_j - x_g)\partial_y W_{gj}V_j & \sum_j (y_j - y_g)\partial_y W_{gj}V_j & \sum_j (z_j - z_g)\partial_y W_{gj}V_j \\ \sum_j \partial_z W_{gj}V_j & \sum_j (x_j - x_g)\partial_z W_{gj}V_j & \sum_j (y_j - y_g)\partial_z W_{gj}V_j & \sum_j (z_j - z_g)\partial_z W_{gj}V_j \end{bmatrix}. \quad (31)$$

The fluid density and stress of the boundary particle denoted by subscript b is then evaluated according to the interpolated values and gradients at ghost nodes using a Taylor series expansion given by,

$$\sigma_b^{\alpha\beta} = \sigma_g^{\alpha\beta} + (x_b - x_g)\partial_x\sigma_g^{\alpha\beta} + (y_b - y_g)\partial_y\sigma_g^{\alpha\beta} + (z_b - z_g)\partial_z\sigma_g^{\alpha\beta}, \quad (32a)$$

$$\rho_b = \rho_g + (x_b - x_g)\partial_x\rho_g + (y_b - y_g)\partial_y\rho_g + (z_b - z_g)\partial_z\rho_g. \quad (32b)$$

To enforce a no-slip boundary condition for the water phase and soil phase (or impermeable boundary for water phase and fully-fixed boundary for soil phase), the boundary particles obtain the velocity of ghost nodes with reversed direction,

$$v_b^\alpha = -v_g^\alpha, \quad (33a)$$

$$q_{wb}^\alpha = -q_{wg}^\alpha, \quad (33b)$$

where the velocity at the ghost nodes is calculated using Shepard corrected summation,

$$v_g^\alpha = \frac{\sum_j v_j^\alpha W_{gj}V_j}{\sum_j W_{gj}V_j}, \quad (34a)$$

$$q_{wg}^\alpha = \frac{\sum_j q_{wj}^\alpha W_{gj}V_j}{\sum_j W_{gj}V_j}. \quad (34b)$$

For the free-slip boundary condition, the off-diagonal components of the stress tensor found at the ghost nodes are reversed to create a frictionless condition along the boundaries as follows,

$$\sigma_b^{\alpha\beta} = \begin{cases} \sigma_g^{\alpha\beta} + (x_b - x_g)\partial_x\sigma_g^{\alpha\beta} + (y_b - y_g)\partial_y\sigma_g^{\alpha\beta} + (z_b - z_g)\partial_z\sigma_g^{\alpha\beta} & \alpha = \beta \\ -\sigma_g^{\alpha\beta} & \alpha \neq \beta \end{cases}, \quad (35)$$

while the fluid density (or pressure) remains the same as those for no-slip boundary using Eq. (32b).

To impose the free-slip condition, the velocities of boundary particles are constructed using,

$$v_b^\alpha = v_g^\alpha - 2v_{g.n}^\alpha, \quad (36a)$$

$$q_{wb}^\alpha = q_{wg}^\alpha - 2q_{wg.n}^\alpha. \quad (36b)$$

The soil density and porosity of boundary particles are set to remain constant to the reference density ρ_0 and reference porosity n_0 during simulation in current application.

3.2.2 Infiltration free-surface boundary condition

For modelling the process of water infiltration in a slope, the infiltration boundary condition needs to be applied to the free surface of the computational domain. The infiltration boundary can be achieved by assigning a prescribed discharge, or pressure, on the infiltration boundaries, or by using a dual boundary condition that enables an automatic change between the discharge-based condition and the pressure-based condition (e.g., Smith et al., 2008). In this study, the second approach is used whereby a zero-pressure boundary condition is applied along the free surface. This approach simulates the case of excessive amount of rainwater that results in ponding and water runoff, resulting in a certain pore water pressure maintained at ground level. It is an extreme scenario occurring during heavy rainfall events, when the rainfall intensity is higher than the saturated hydraulic conductivity (Martinelli et al., 2020), and the suction pressure at the ground surface become zero. Examples of the application of this type of boundary condition for landslide failures are studied by Cuomo and Della Sala (2013), Wang et al. (2018), Lee et al. (2019), and Lei et al. (2020). However, it will be demonstrated later that with the present SPH formulations, zero-pressure boundary conditions are implicitly satisfied at the free surface (i.e., natural boundary condition) and there is no requirement of applying zero-pressure to the free-surface particles (e.g., Wang et al., 2018) or implementing a surface boundary algorithm (e.g., Bandara et al., 2016), which simplifies the boundary treatment.

In addition, periodic boundary condition (PBC) is also used in this study (Domínguez et al., 2021) for the simulation of 1-D infiltration test and Liakopoulos's drainage test to produce a uniform seepage flow, as described in Section 4. The PBC is achieved by allowing particles near the lateral open boundary to interact with the particles near the complimentary lateral open boundary on the other side.

3.3. Initial conditions

Reasonable initial conditions are required to achieve reliable and accurate results. In this study, the initial stress distributions are generated by applying gravity to the soil domain with the given pore pressure distribution, in which the suction pressure is calculated from the initial water content using the soil–water retention properties. To achieve fast stabilization in the stress field, a damping term is introduced into the momentum equation (e.g., Bui and Fukagawa, 2013), according to,

$$\left\langle\frac{\mathrm{d}v^\alpha}{\mathrm{d}t}\right\rangle_i = \sum_{j=1}^{N}\frac{m_{\mathrm{mj}}}{\rho_{mi}\rho_{\mathrm{mj}}}\left(\sigma_i^{\prime\alpha\beta} + \sigma_j^{\prime\alpha\beta} + \Pi_{ij}\delta^{\alpha\beta}\right)\frac{\partial W_{ij}}{\partial x_i^\beta} - \sum_{j=1}^{N}\frac{m_{\mathrm{mj}}}{\rho_{mi}\rho_{\mathrm{mj}}}\left(S_i p_i + S_j p_j\right)\frac{\partial W_{ij}}{\partial x_i^\alpha} + g_i^\alpha + d_i^\alpha, \quad (37)$$

where d is the damping force term given by,

$$d_i^\alpha = -\xi\sqrt{\frac{E}{\rho h^2}}v_i^\alpha, \quad (38)$$

in which, ξ, is the damping coefficient, and its value is commonly taken as 0.02 (Bui and Fukagawa, 2013).

Once the state of equilibrium is achieved and the initial conditions are obtained, the damping term is "switched off" and the water infiltration is allowed.

3.4. Stress diffusive term

One of the key challenges of the application of SPH, or other particle methods such as MPM, to geomechanics problems is the presence of

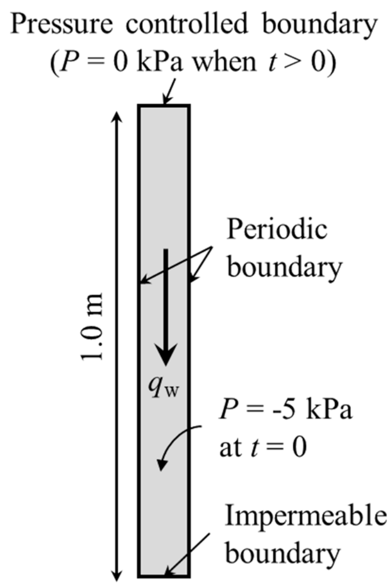

Fig. 4. Numerical configuration and boundary conditions used in the infiltration test case.

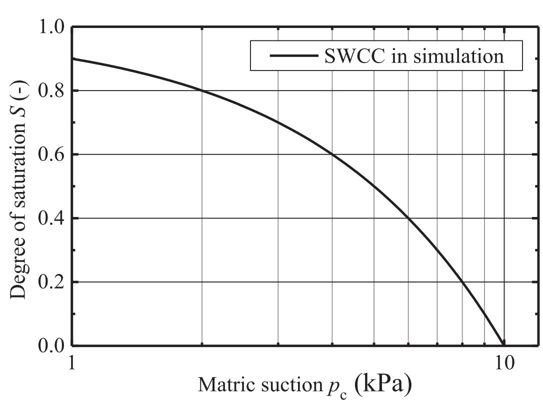

Fig. 5. SWCC adopted in the numerical simulation of infiltration test.

numerical oscillations and noise in the stress field when large deformations are achieved, which may affect the accuracy of the numerical analysis. One simple way to improve the spurious stress field is to adopt the stress diffusion algorithm proposed by Feng et al., (2021), which requires including a diffusion term in the stress updating, taking form as,

$$\frac{D\sigma_i^{\alpha\beta}}{Dt} = D^{ep}\dot{\varepsilon}_i^{\alpha\beta} + D_i^{\alpha\beta}, \tag{39}$$

where D^{ep} is the elasto-plastic stiffness matrix and $D_i^{\alpha\beta}$ is the stress diffusion term defined for each stress component as,

$$D_i^{\alpha\beta} = 2\zeta h c_0 \sum_j \psi_{ij}^{\alpha\beta} \frac{\boldsymbol{x}_{ij}\cdot\nabla_i W_{ij}}{|\boldsymbol{x}_{ij}|^2 + 0.01h^2} \frac{m_{mj}}{\rho_{mj}}, \tag{40}$$

where ζ is the diffusion coefficient and normally take values as 0.1. The parameter ψ_{ji} is the diffusion operator that has the dimension of the physical quantities to be diffused.

The idea behind Eq. (40) is to approximate the Laplacian of stress to achieve a smooth stress profile. The stress diffusion operator should be constructed to restore the global convergence and consistency over the whole domain (Antuono et al., 2012). Feng et al. (2021) proposed a stress diffusion operator modified from Fourtakas et al. (2019), and it was shown that the formulated diffusion operator is able to smooth out the stress profile with good accuracy and convergence. In order to adopt the modified diffusion operator, the stress gradient at geostatic condition is required for restoring consistency at the free-surface as shown by Feng et al. (2021). This is convenient for a single-phase problem or total stress analysis in which the stress is the only state variable, but for a two-phase coupled problem, as the stress profile of unsaturated soils is influenced by the soil hydraulics and hydrological conditions, it is difficult to obtain a unique solution of stress gradient in static condition for the stress diffusion operator. In this study, following the method presented by Antuono et al. (2010), a general type of diffusion term is formulated for the purpose of diffusing stress for the coupled hydro-mechanical analysis, which reads,

$$\psi_{ij}^{\alpha\beta} = \left(\sigma_i^{\alpha\beta} - \sigma_j^{\alpha\beta}\right) - \frac{1}{2}\left(\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_j^L + \left\langle\nabla\sigma^{\alpha\beta}\right\rangle_i^L\right)\cdot\boldsymbol{x}_{ij}, \tag{41}$$

where $\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_j^L$ and $\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_i^L$ are the renormalised stress gradients defined as,

$$\left\langle\nabla\sigma^{\alpha\beta}\right\rangle_j^L = \sum_j \left(\sigma_j^{\alpha\beta} - \sigma_i^{\alpha\beta}\right)\boldsymbol{L}_i \nabla_i W_{ij}\frac{m_{mj}}{\rho_{mj}}, \tag{42a}$$

$$\boldsymbol{L}_i = \left(\sum_j \left(\boldsymbol{x}_j - \boldsymbol{x}_i\right) \otimes \nabla_i W_{ij}\frac{m_{mj}}{\rho_{mj}}\right)^{-1}. \tag{42b}$$

3.5. Time integration scheme

The stability of a numerical algorithm is dependent of the time integration scheme. In the present work, the time stepping scheme is an explicit second-order predictor–corrector scheme (Crespo et al., 2015). This scheme predicts the evolution in time at the middle of the time step (i.e., predictor step); the obtained values are then corrected using the updated forces at half time steps (i.e., corrector step), followed by the computation of the values at the end of the time step.

However, explicit time integration schemes are conditionally stable. Commonly, the Courant–Friedrichs–Lewy (CFL) stability condition is introduced to prescribe the maximum allowable time step, in which the critical time step needs to be less than the time for a compression wave to travel through the characteristic length of the discretized system, but for the two-phase problem, the influence of permeability K needs also to be accounted for (Mieremet et al., 2016). In this study, the length of time step is restricted by the CFL condition, the maximum force term, the numerical speed of sound, and the diffusion of seepage flow. The critical time step is computed as (Feng et al., 2021, Lian et al., 2021),

$$\Delta t_f = \min_i\left(\sqrt{h/|\boldsymbol{f}_i|}\right), \tag{43a}$$

$$\Delta t_{cv} = \min_i \frac{h}{c_s + \max_j\left|\frac{h v_{ij}^{\alpha} x_{ij}^{\alpha}}{x_{ij}^{\alpha} x_{ij}^{\alpha} + \eta^2}\right|}, \tag{43b}$$

$$\Delta t_{sp} = \min_i\left(\gamma_w \frac{h^2 c_v}{2K}\right), \tag{43c}$$

$$\Delta t = C_0 \min\left(\Delta t_{cv}, \Delta t_f, \Delta t_{sp}\right), \tag{43d}$$

where C_0 is the Courant number,

$$c_v = n\left(\frac{S}{K_w} + \frac{\partial S}{\partial p}\right). \tag{44}$$

The implementation of the fully coupled hydro-mechanical model described above is achieved in the CPU branch of open-source SPH code DualSPHysics (Domínguez et al., 2021).

4. Validation cases

The validity of the numerical scheme is examined in this section

Table 1
Material and hydraulic properties for the 1-D infiltration test.

Paramters	Unit	Value
h/dp	–	1.8
Courant number	–	0.2
Soil density ρ_s	kg/m^3	2100
Water density ρ_w	kg/m^3	1000
Porosity n	–	0.3
Water bulk modulus K_w	Pa	8.0×10^6
Hydraulic conductivity K	m/s	5.0×10^{-3}
SWCC constant a_v	Pa^{-1}	1.0×10^{-4}

using an infiltration test and a drainage test of a soil column. Numerical results are compared against available analytical solutions and experimental data.

4.1. Infiltration test

The infiltration test is used to examine the infiltration boundary condition and the ability of the proposed scheme to solve the transient unsaturated seepage problem. More specifically, the infiltration test simulates the water infiltration along the top surface of a rigid unsaturated soil column (see Fig. 4). This test case has been used for validation purposes by Yerro (2015) and Lei et al. (2020) since an analytical solution is available for comparison.

The geometries and boundary conditions of the infiltration test are presented in Fig. 4. The soil column is assumed to be initially unsaturated with a constant suction pressure of $p_c = 5$ kPa. Periodic boundary conditions are applied to both sides to achieve one-dimensional water flow. The bottom boundary of soil column is impermeable, while the zero-pressure boundary condition is implicitly satisfied at the top. Gravity is not considered in this application. Once the seepage flow is allowed, the wetting process starts at the top of the soil column. To obtain an analytical solution, the permeability is assumed to be constant. The linearized SWCC model expressed in Eq. (23a) is used, taking $S_{max} = 1.0$ and $S_{min} = 0.0$. The illustrative diagram of the adopted SWCC model is shown in Fig. 5.

The 1-D vertical water infiltration within a rigid unsaturated soil column can be mathematically expressed by,

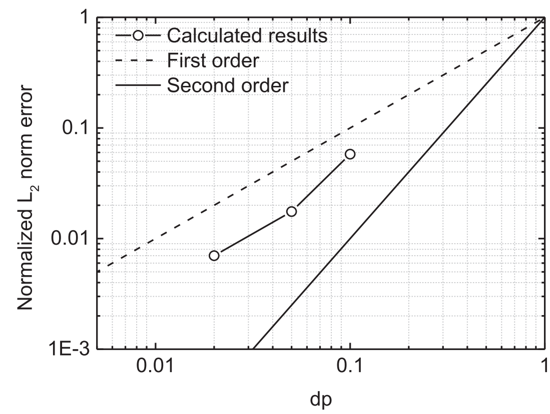

Fig. 7. Convergence plot for the infiltration test: the normalised L_2 norm error of the suction pressure at 10 s.

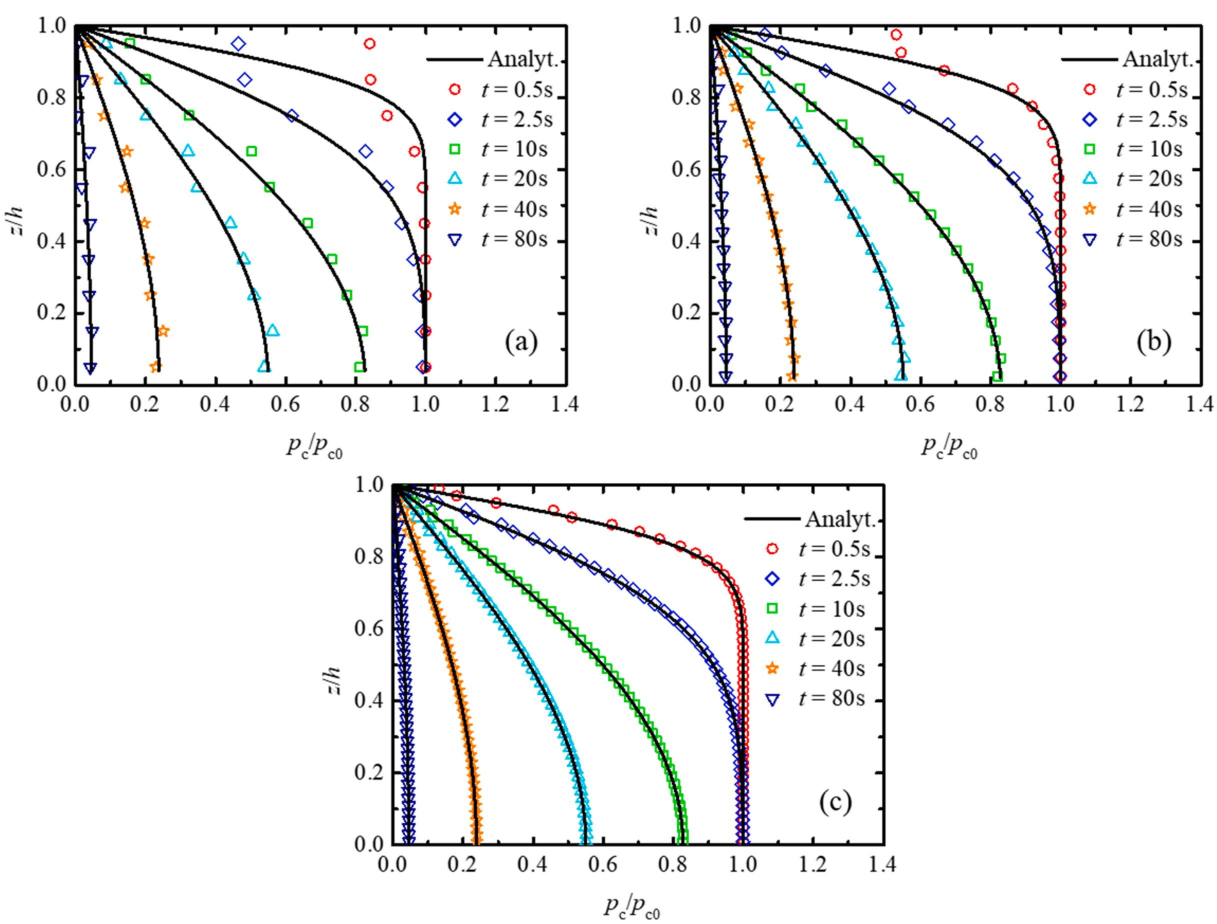

Fig. 6. Evolution of suction pressure along depth z (a) $dp = 0.1$, (b) $dp = 0.05$, and (c) $dp = 0.02$.

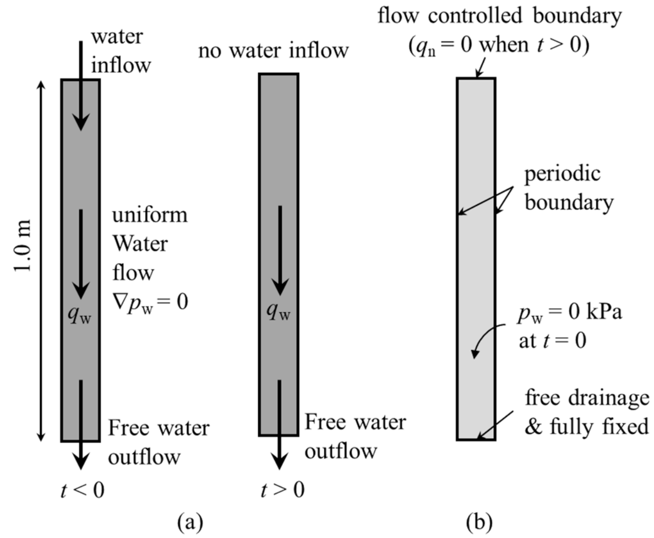

Fig. 8. Drainage test: schematic diagram of (a) the experimental configuration and (b) the numerical configuration.

Table 2
Material properties and numerical parameters for the Liakopoulos' test.

Paramters	Unit	Value
h/dp	–	1.3
Courant number	–	0.2
Artificial viscosity α	–	0.1
Soil density ρ_s	kg/m^3	2000
Young's modulus E	Pa	1.3×10^6
Poisson's ratio ν	–	0.4
Water density ρ_w	kg/m^3	1000
Porosity n	–	0.2975
Water bulk modulus K_w	Pa	2.0×10^7
Hydraulic conductivity K	m/s	4.41×10^{-6}

$$\frac{\partial p}{\partial t} = C_i \frac{\partial^2 p}{\partial z^2}, \tag{45}$$

where z is the distance in the infiltration direction and the coefficient C_i can be written as,

$$C_i = \frac{K}{n\gamma_w a_v}. \tag{46}$$

Considering the initial and boundary conditions, given by,

$$\begin{cases} p(z,t)\,|_{t=0} = p_c \\ p(z,t)\,|_{z=h} = 0 \\ \partial p(z,t)/\partial z\,|_{z=0} = 0 \end{cases}, \tag{47}$$

the analytical solution for Eq. (45) is equivalent to the solution of 1-D consolidation problem in saturated porous media, known as the Terzaghi's solution. The dimensionless time T used for the analytical solution is defined in the conventional way as,

$$T = \frac{C_i t}{h^2}. \tag{48}$$

The model parameters used in the simulation are listed in Table 1. Three sets of particle resolutions of 0.1, 0.05 and 0.02 m are adopted, resulting in 10, 20 and 50 particles along the column height, respectively.

Fig. 6 shows the comparison of the suction pressure evolution against the analytical solutions for three particle resolutions, i.e., $dp = 0.1$, 0.05 and 0.02 m. Agreements between numerical and analytical results are achieved with the L_2 norm error less than 5 %. The calculated suction profile fluctuates when a large value of dp is used, resulting in slight deviations from the analytical solution near the top of the soil column. This can be attributed to the local inconsistency near the free-surface of the enforcement the zero-pressure boundary condition, which become noticeable for a coarse resolution. For smaller dp, the solution converges towards the analytical solution.

Fig. 7 shows the results from a convergence study on the suction pressure, from which it can be seen that a satisfactory order of convergence of 1.604 is achieved.

4.2. Liakopoulos's drainage test

The drainage test described by Liakopoulos (1964) is used to examine the validity of the proposed scheme for the coupling of unsaturated seepage flow and soil deformation. This case refers to the desaturation process of an initially fully saturated soil column due to gravity, and has been investigated by Bandara et al. (2016), Wang et al. (2018) and Lei et al. (2020) to verify the implementation of coupled hydro-mechanical frameworks for unsaturated soils.

Fig. 8(a) shows a schematic diagram of the Liakopoulos' test. Before the experiment starts, a uniform water flow condition with zero pressure gradient is achieved by adding a constant water inflow at the top of the specimen, after which the water inflow at top is stopped and the bottom drainage is opened to initiate the test. The moisture tension along the height was monitored during the experiments.

Fig. 8(b) presents the initial and boundary conditions applied in the simulation. The initial suction pressure is set as zero across the domain

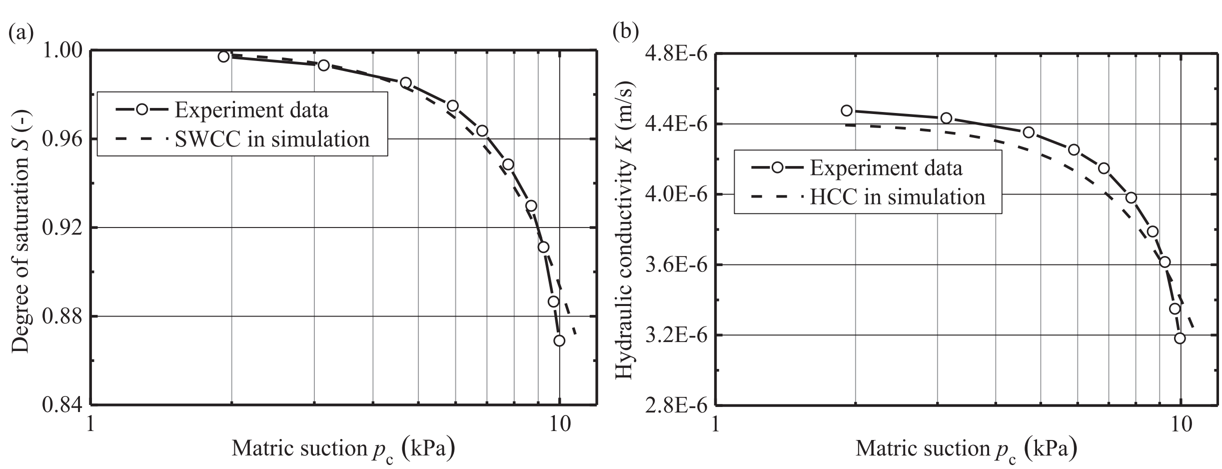

Fig. 9. Comparisons of the hydraulic model used in simulation against the experiment data by Liakopoulos (1964): (a) SWCC and (b) HCC.

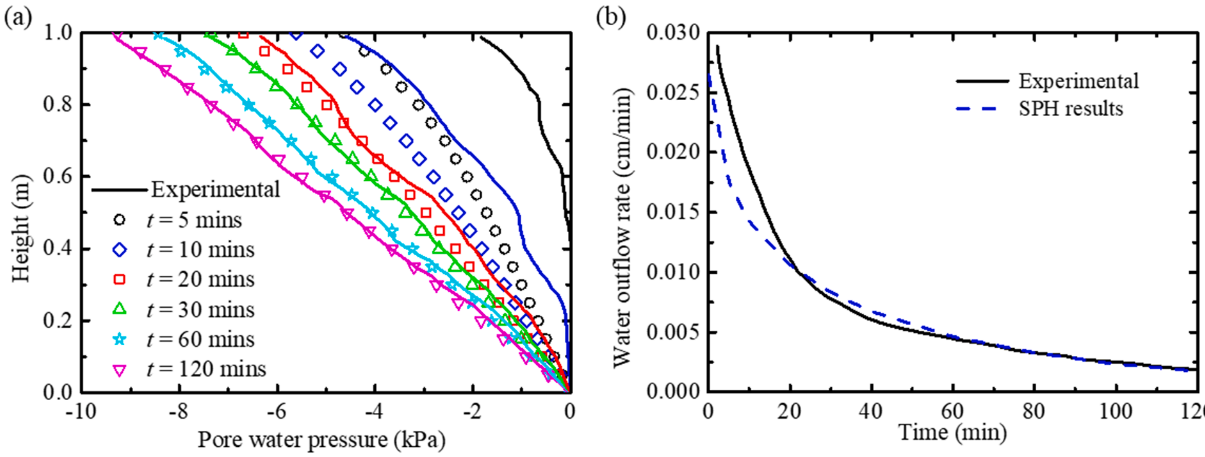

Fig. 10. Comparisons between SPH results and experimental data: (a) pore water pressure with height (b) water outflow rate at the bottom.

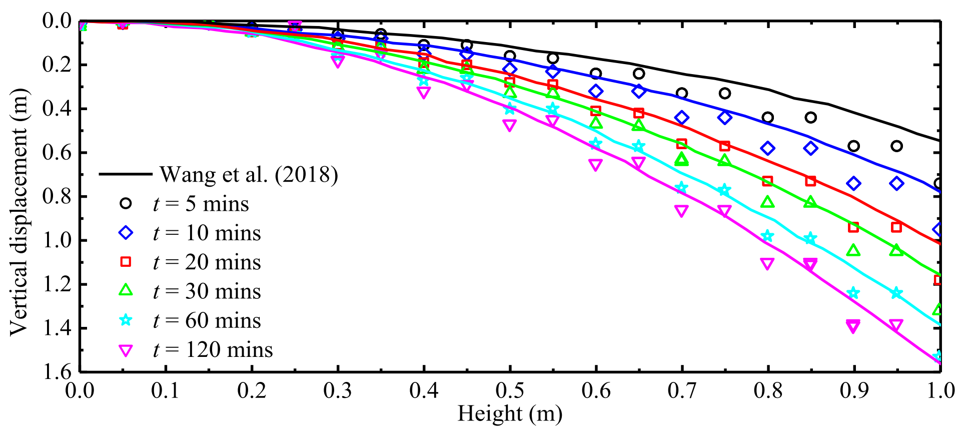

Fig. 11. Comparison between predicted and experimental vertical displacement for the Liakopoulos's drainage test.

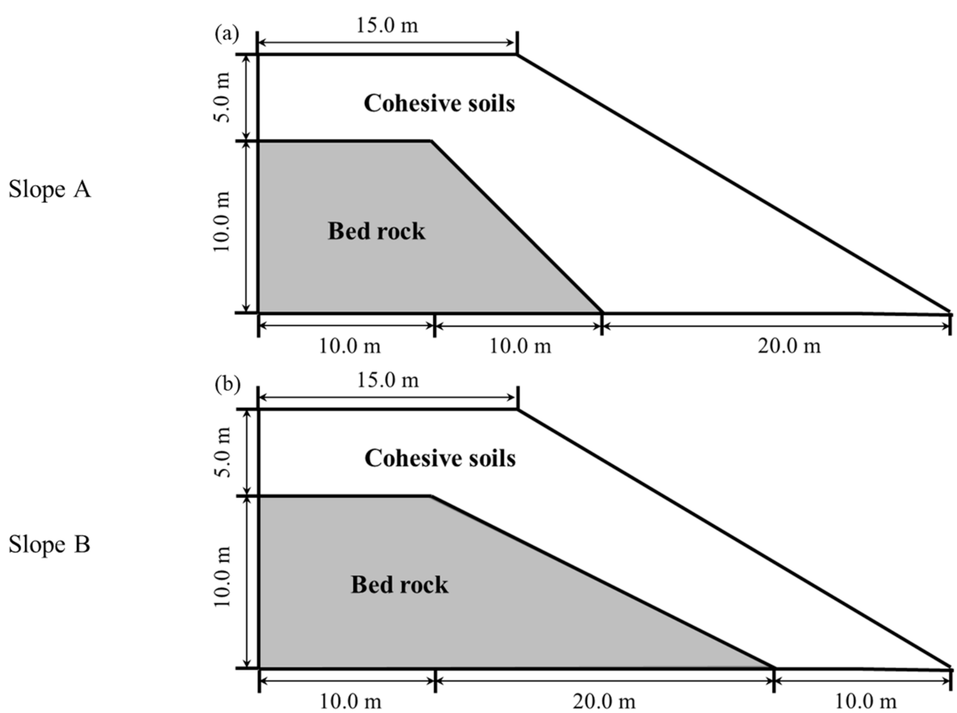

Fig. 12. Illustrative diagram of the slope models: (a) Slope A with a deep soil cover and (b) Slope B with a shallow soil cover.

Table 3
Material and hydraulic properties for the slope failure test.

Paramters	Unit	Value
h/dp	–	1.3
Courant number	–	0.2
Artificial viscosity α	–	0.1
Soil density ρ_s	kg/m^3	2650
Young's modulus E	MPa	10.0
Poisson's ratio ν	–	0.3
Friction angle ϕ	°	21.0
Peak cohesion c_p	kPa	10.0
Residual cohesion c_r	kPa	3.0
Softening coefficient η_c	–	5.0
Water density ρ_w	kg/m^3	1000
Porosity n	–	0.3
Water bulk modulus K_w	Pa	3.0×10^7
SWCC constant α_v	Pa^{-1}	1.0×10^{-5}
Hydraulic conductivity K	m/s	1.0×10^{-3}

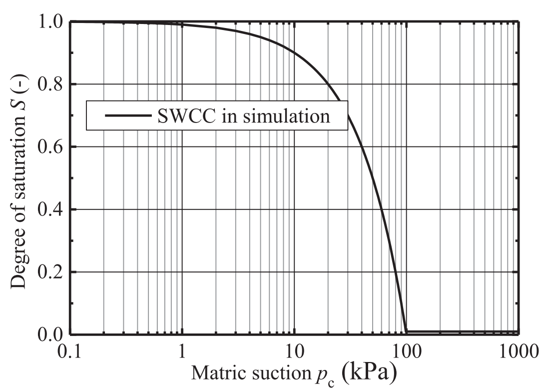

Fig. 13. SWCC adopted in the simulation of rainfall-induced slope failure.

(i.e., fully saturated). The top of the soil column is set as zero normal-flux (or impermeable), while the zero-pressure boundary condition is applied to the bottom. The initial stress profile and particle positions are obtained by applying the gravitational field to the soil column with the use of damping term. Numerical experiments show that the damping process helps to reduce the initial oscillations and errors in the prediction of particles' displacements.

A linear elastic model is adopted to simulate the soil behaviour. The relationship between saturation and suction pressure, saturation and relative permeability are taken as (Schrefler and Scotta, 2001),

$$S = 1.0 - 0.10152\left(\frac{p_c}{\rho_w g}\right)^{2.4279}, \tag{49a}$$

$$K_{rel} = 1.0 - 2.207(1.0 - S)^{1.0121}. \tag{49b}$$

The material and hydraulic properties are listed in Table 2, which are taken from Schrefler and Scotta (2001), and Bandara et al. (2016). The comparisons of the fitted SWCC and HCC against experiment data are plotted in Fig. 9. The initial particle spacings are set to 0.05 m, resulting in 20 particles over the height of the domain.

Fig. 10 compares the experimental data and SPH results. The evolutions of pore water pressure along the column are presented in Fig. 10 (a). It can be seen that the desaturation process results in a gradual increase of suction pressure along the column. For the first 10 mins, the calculated results overestimate the build-up of the suction pressure, which is consistent with results observed in previous FEM studies (Schrefler and Scotta 2001) and the MPM analysis (Bandara et al., 2016); while satisfactory agreement with experimental results is achieved after about 20 mins of physical time. Fig. 10(b) shows the water outflow rate recorded at the bottom, which shows a decreasing trend with time caused by the corresponding decrease in the rate of change of pore water pressure. It can be noted that the predicted water outflow rates are in overall agreement with experimental observations.

The predicted vertical displacement profiles along the height of the column are plotted in Fig. 11; results from Wang et al., (2018) are also plotted on the same figure for comparison. Note that the effective stresses are modified during the drainage as the suction pressure increases. The deformation of the soil skeleton due to the changes in the effective stress accumulates with time, resulting in the settlement profile of the soil column shown in Fig. 11. Although slight fluctuations are observed in the settlement profile, the computed results are in agreement with those reported by Wang et al., (2018), confirming the applicability of the present framework for the treatment of transient problems in unsaturated soils.

5. Application example: slope failure due to rainfall infiltration

The proposed numerical scheme is used to simulate an initially unsaturated slope that becomes unstable due to water infiltration, for example, caused by a heavy rainfall event. The case study aims at predicting both the onset of failure as well as the post-failure deformation. Schematic diagrams of the slope geometries are plotted in Fig. 12. It is noted that all slopes have the same heights (15 m), widths (15 m) and lengths (40 m) and inclination of 1:1.67. To examine the influence of the soil cover depth on the post-failure motion of the landslides, two slope geometries with bedrock inclinations of 1:2 and 1:1 are considered; these slope models are hereafter referred to as slope A and slope B.

The bedrock is modelled as an impermeable material, using the solid wall boundary condition; no-slip boundaries are adopted to simulate a perfectly rough interaction between bedrock and the above soil layer. Free-slip conditions are used to the left sides of the domain to enforce frictionless conditions. The water infiltration initiates once the seepage flow is allowed in the computational domain thanks to the dynamic free-surface boundary condition.

The soil behaviour is simulated using the elastic–plastic Drucker-Prager model with strain-softening. The linear SWCC model is used to characterise the hydraulic properties of the soils. Material and hydraulic properties are listed in Table 3.

The adopted SWCC is plotted in Fig. 13. It is found that, with the adopted linear SWCC model, air will significantly penetrate the soil at $p_c = 1$ kPa while the water content reaches its residual value at $p_c = 100$ kPa. Similar trends can be found in the literature for silty soils as shown in Stange and Horne (2005). Nevertheless, to reduce the computational time of the simulation, the selected hydraulic conductivity may be at the higher end of the typical values for silty soils. This approximation speeds up the wetting process and allows our model to be tested until very large deformations are achieved in the sliding mass without demanding computational costs.

The initial particle spacings of 0.5 m, 0.2 m, and 0.1 m are adopted, resulting a total of 1037, 6524, and 26,175 particles for slope A and 835, 5275, and 21,175 particles for slope B. The value of dp adopted in this case is larger than those used in the element tests as shown in Section 4. It is noted that a larger dimension of the physical domain is involved in this case. Since the proposed numerical scheme is converging, the selection of a relatively coarse resolution enables both an acceptable computational cost while ensuring the computation error of the numerical solution is within a predefined range. The slope is assumed to be initially unsaturated, with the initial ground water surface located at the bottom of the slope. The initial distribution of pore water pressure is set as hydrostatic, while the initial stress profile is generated by applying a gravitational field and using the damping term to achieve the fast equilibrium. Water infiltration is then allowed to initiate the simulations. In the following, the results from the medium particle resolution ($dp = 0.2$ m) are presented.

Fig. 14 shows the temporal evolution of the accumulated deviatoric plastic strain and pore water pressure in slope A. It can be seen that the failure starts at a time $t = 101$ s near the slope's toe, where the moisture

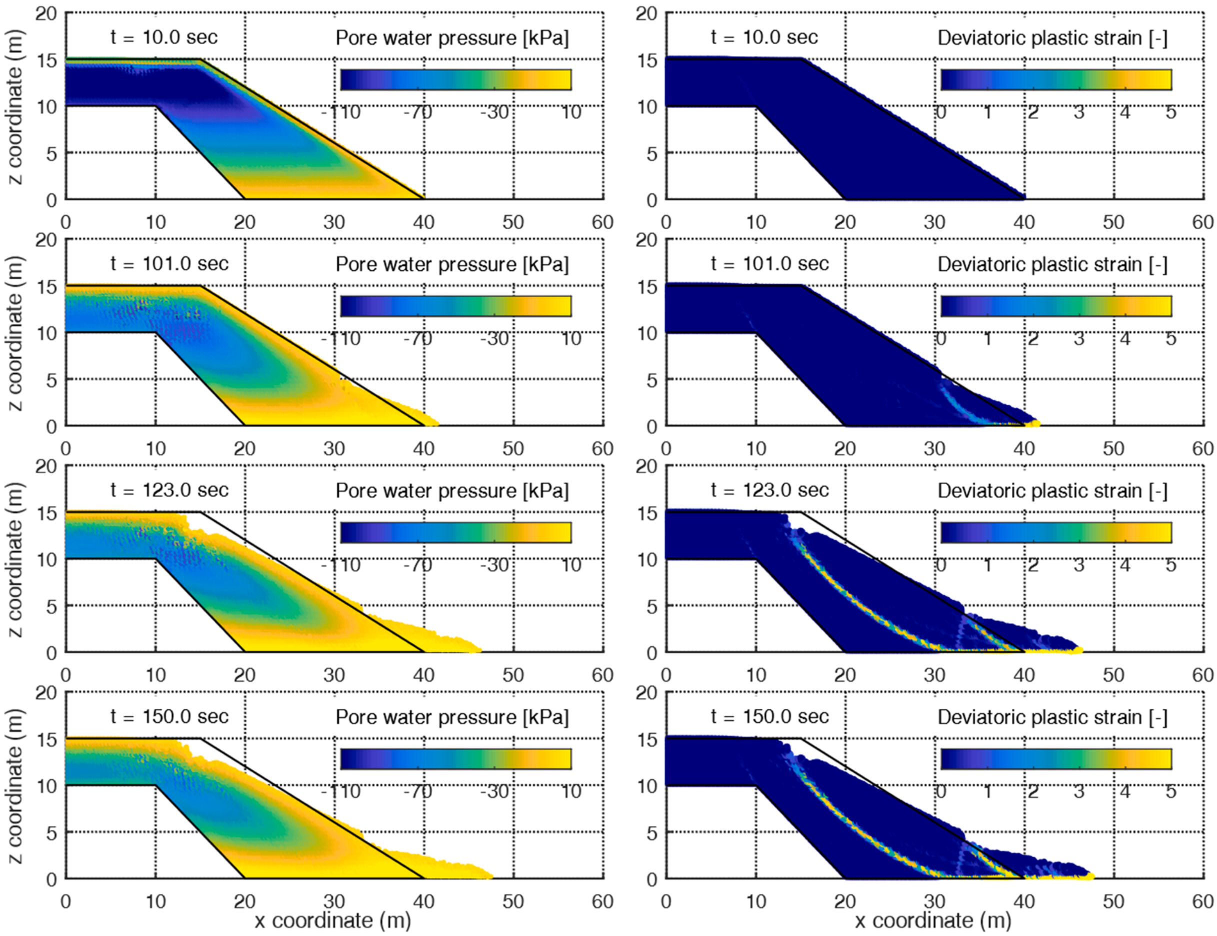

Fig. 14. Pore pressure contour (left column) and accumulated deviatoric plastic strain contour (right column) at different time instants for slope A.

content first reaches the saturation condition while the remaining soil is still unsaturated. Due to the loss of the support at the slope toe, and the continuous reduction in soil strength (softening) due to water infiltration, a successive concave-upward circular failure surface forms at $t =$ 123 s. This retrogressive-type of failure pattern is commonly observed in the real field (Orense et al., 2004, Tohari et al., 2007, Regmi et al., 2014, Chueasamat et al., 2018) as the instability developed at the slope's toe influences the stability of the upper portion of the slope, resulting in instabilities of relatively large volume of soil.

In Fig. 14, the evolution of pore water pressure due to heavy rainfall is clearly reproduced with the adopted free-surface boundary condition. Saturated wetting fronts are advancing from the ground surface at the start of simulation, and the toe of the slope is already saturated when the first slide initiates at t = 101 s. The wetting front continuously progresses from the failure region towards the centre of the slope, while the suction pressure dissipates simultaneously, contributing to the failure propagation.

The calculated results of vertical and shear effective stress profiles obtained with and without the diffusion term are shown in Fig. 15 and Fig. 16, respectively. It is found that without the use of diffusion term, numerical noise and fluctuations develop within the simulation domain, which is a common issue in other particle-based methods, including MPM and SPH. However, these deficiencies can be much improved by introducing numerical diffusion in stress, resulting in a noise-free smooth stress distribution.

Fig. 16 shows that, at the start of the simulation, the maximum shear stresses develop at the interface between the slope and the underlying boundary, providing resistance against horizontal slip. Before the onset of failure, there is a rapid development of shear stresses above and below the shear plane. At the onset of the instability these shear stresses attain their maximum values, before decreasing as the sliding mass moves downslope.

Another test scenario is investigated using the geometry of slope B, with a shallower soil cover compared to that in slope A. The failure behaviour of slope B is presented in Fig. 17 using the contour plots of accumulated deviatoric plastic strain and pore water pressure at different time instants. Evidently, a different failure mechanism can be observed due to the presence of a shallow soil cover. The slope remains stable for 79 s, after which relatively large plastic strains develop at the toe of the slope. This is followed by the rapid formation of the integral failure across the main slope. Once the failure surface is well developed, the soils above the failure surface becomes unstable and starts to move downslope along the failure surface. With the gradual movement of the sliding mass, the second failure develops within the moving mass at $t =$ 99 s. The failure propagates further downwards the slope toe at $t = 150$ s, in which a third sliding surface is observed. This type of failure mechanism is consistent with the experimental observations by Regmi et al. (2014) for a shallow slope subjected to rainfall.

Unlike slope A, where a global stability is maintained for a certain time after the local failure at the toe and before the initiation of the global slide, for slope B the global slide across the main slope develops at the onset of instability without any observation of local failure. This finding may indicate that the onset of integral failure in shallow-seated slopes may give fewer warning signs than in deep-seated slopes. Another difference in failure mechanism is the shape of the global sliding surface. In slope A, it is a continuous circular arc, which is a typical type of failure surface in homogeneous materials; the failure surface of slope B involved a combination of curved and planar slip surfaces. The presence

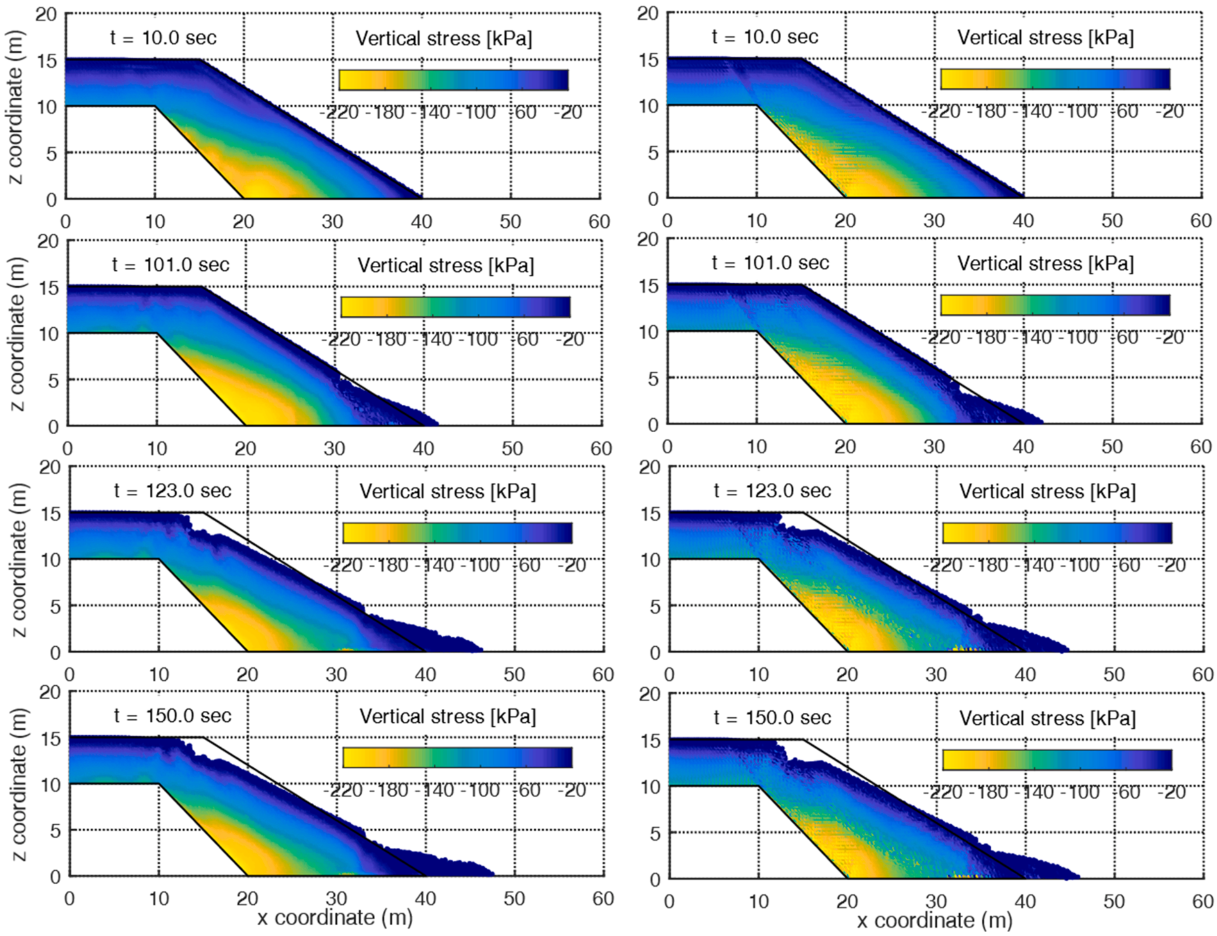

Fig. 15. Vertical effective stress contour at different time instants for slope A with the diffusion term (left column) and without the diffusion term (right column).

of a shallow bedrock which limits the propagation of the sliding surface to greater depths while forcing the sliding mass to move parallel to the impermeable ground along a planar slip surface. As a consequence, the motion of sliding mass for slope A is purely rotational while for slope B it involves both translational and rotational slips.

It is well known that the accuracy of SPH interpolation is dependent of the particle spacing (Quinlan et al., 2006). With the refinement of *dp*, the SPH error is reduced until the discretisation error dominates. The influence of spatial discretisation on the post-failure motion is also investigated and the results at $t = 120$ s are presented in Fig. 18. The locations of main shear band are similar for the three simulations but more sub-slides can be found at the slope toe when decreasing *dp*. This observation has also been reported by Bui and Nguyen (2021) for the case of retrogressive failures of slopes in sensitive clays. The discrepancies in the failure surfaces can be attributed to the accumulation of numerical errors, which may be become significant at large deformations. As the particle resolution is refined, the magnitude of the error from SPH interpolation is reduced, thus providing the ability to capture more broken blocks, which may not be able to be reflected with a large *dp*.

6. Discussion

Rainfall-induced slope instability involves the development of continuous sliding surfaces at the pre-failure stage, and the subsequent motion of sliding masses at the post-failure stage. The integrated analysis of the failure and post-failure behaviour, which is challenging for traditional mesh-based methods, has been implemented in this study using a new single-layer multi-phase coupled SPH model. The proposed method enables the simulation of water infiltration at the free surface without a surface searching algorithm and it avoids the problem that boundary conditions do not align well with the boundary (e.g., the MPM model by Wang et al. 2018). It is shown that the stress field in present analysis is smooth at all stages of the failure process, even at large deformations. However, the proposed formulations still present a number of limitations due to the assumptions made and simplifications applied. They are discussed in the following together with suggestions for future works.

The method adopted the natural zero-pressure free-surface boundary condition to simulate the water infiltration process. This boundary treatment simplifies the algorithm since it avoids the complex surface searching algorithm (e.g., Bandara et al., 2016, Ceccato et al., 2021) but it assumes a fully saturated condition along the free surface and saturated wetting fronts advancing from the ground surface. It may result in an extreme infiltration rate that could lead to conservative failure prediction. However, in order to enforce a prescribed infiltration rate along the free surface to predict slope instability with a given rainfall data or to investigate the influence of infiltration parameters (e.g., rainfall intensity and duration) on the failure mechanism of the slope, additional development is required to identify free-surface particles for SPH. It is also noted that, with the use of discharge-based condition, ponding may take place if the inflow rate exceeds the storage capacity of the soil. In such a case, a new algorithm is required to enable a switch between discharge-based condition and pressure-based condition in SPH. Future developments are required for a more advanced infiltration boundary condition in SPH.

The present numerical frameworks are formulated based on a generalised elastic–plastic soil model with the assumption of a constant

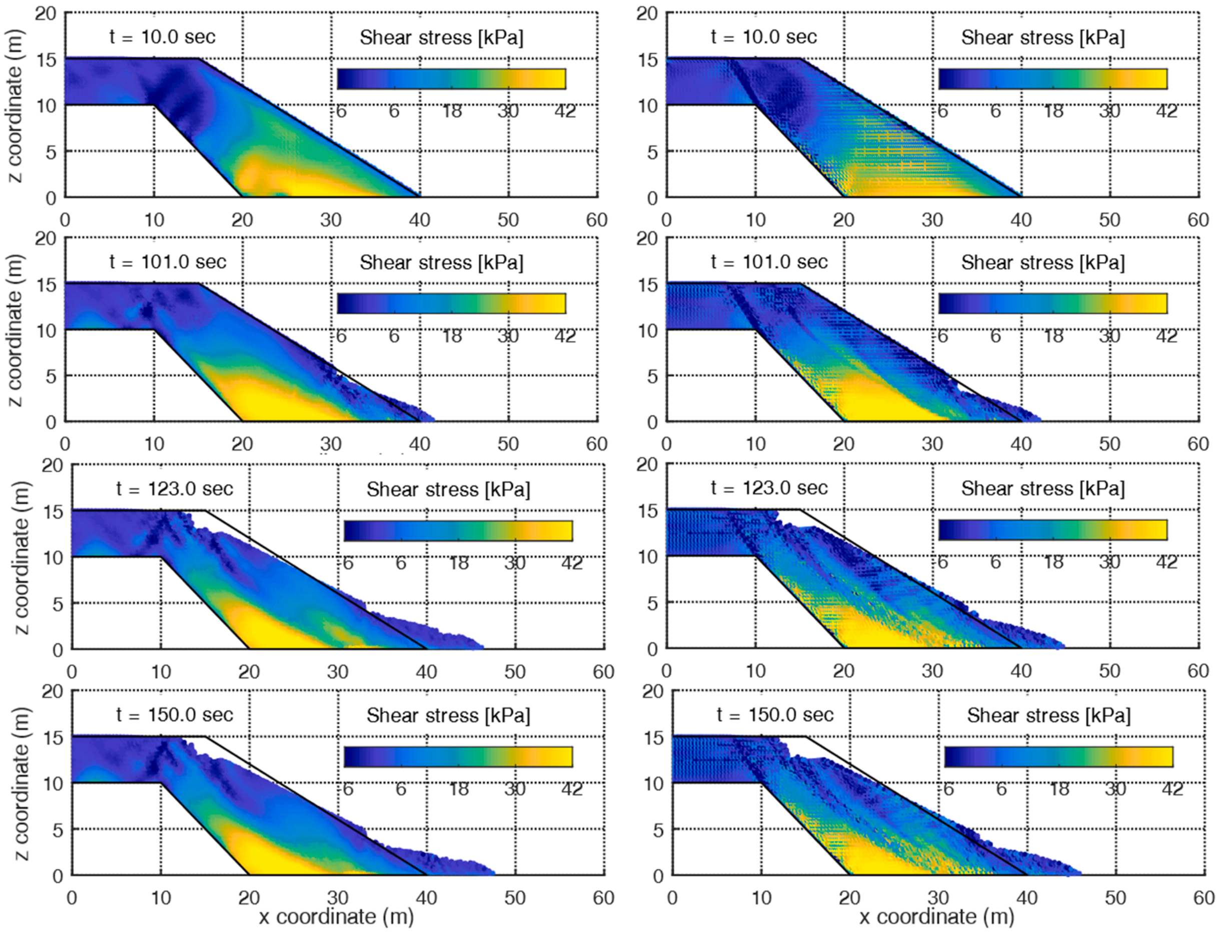

Fig. 16. Shear effective stress contour at different time instants for slope A with the diffusion term (left column) and without the diffusion term (right column).

air pressure. To simulate the behaviour of unsaturated soils, the SWCC model is combined with a strain-softening Drucker-Prager soil model that is formulated using Bishop's effective stress with the effective stress parameter χ equal to the degree of saturation S (e.g., Bandara et al., 2016, Wang et al., 2018, Lei et al., 2020, Ceccato et al., 2021). It is noted that this type of effective stress reduces to Terzaghi's effective stress at fully saturated condition, providing a smooth transition of stress state between the unsaturated and saturated condition. The dependency of soil strength on suction is taken into account since the evolution of effective stress with suction is linked to the SWCC model. This numerical configuration provides a simple but realistic representation of unsaturated soil behaviour and the common features of rainfall-induced landslides are captured as shown in the present work. Nevertheless, to account for other features relevant to unsaturated soils, such as the wetting induced volume change or collapse that result in initial failures (Gens, 2010), the proposed formulation requires the implementation of more advanced unsaturated soil constitutive models, such as those that incorporate SWCC within the constitutive model (Wheeler et al., 2003), or those that adopt more sophisticated stress variables (Sheng et al., 2008). An overview of the constitutive modelling of unsaturated soils can be found in D'Onza et al. (2011). In addition, the proposed formulations can be further extended to consider the air pressure variation by including the momentum/mass balance laws of air phase, for the case that air pressure plays a role.

Another limitation of the present model that will be addressed in future work are extending the implementation to 3-D using hardware acceleration.

Despite the limitations mentioned above, the proposed formulation provides a new route for the analysis of large deformation of unsaturated soils. Its potential applications are broader, going beyond the test case of rainfall-induced landslides presented here. For example, it can be applied to the study of seepage flow through a dam where the boundary conditions defining the hydraulic head, seepage front are correctly implemented.

7. Conclusions

This paper has presented a new two-phase single-layer SPH formulation for unsaturated soils to study the infiltration-induced slope collapse. A new extension of the first-order consistent wall boundary treatment by Feng et al. (2021) has been proposed for the coupled problem to enforce no-slip/free-slip boundary conditions in fluid/solid phase. It is found that the proposed formulations provide an implicit implementation of zero-pressure boundary condition at the free-surface. This surface boundary condition avoids the use of complex particle searching algorithms (e.g., Bandara et al. 2016, Ceccato et al. 2021), and can be conveniently adopted to simulate water infiltration problems, such as the ones described in the present manuscript. A new formulation of stress diffusion term is presented to reduce numerical noise in the stress field at large deformations. Comparing to the formulation by Feng et al. (2021), which is limited to gravity-dominated failure, the new method is general and can be adopted for any scenarios, which is well-suited for the coupled problem. The proposed model is validated for two tests cases, representing the infiltration test and the drainage test of a soil column.

Finally, the proposed model is employed for the analysis of rainfall-induced slope collapse. The analysis results show that the geometries of bedrock or the depth of slope play an important role in the failure

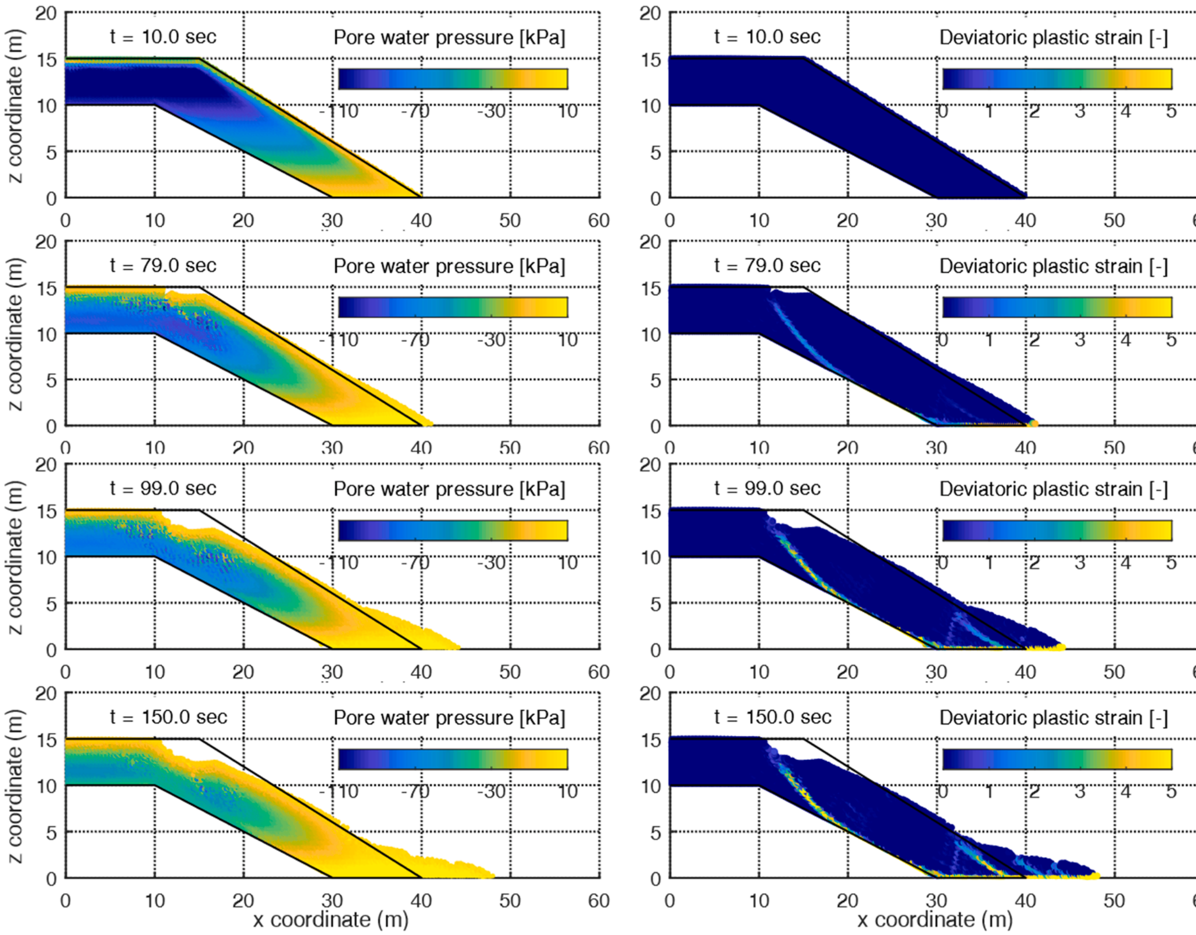

Fig. 17. Pore pressure contour (left column) and accumulated deviatoric plastic strain contour (right column) at different time instants for slope B.

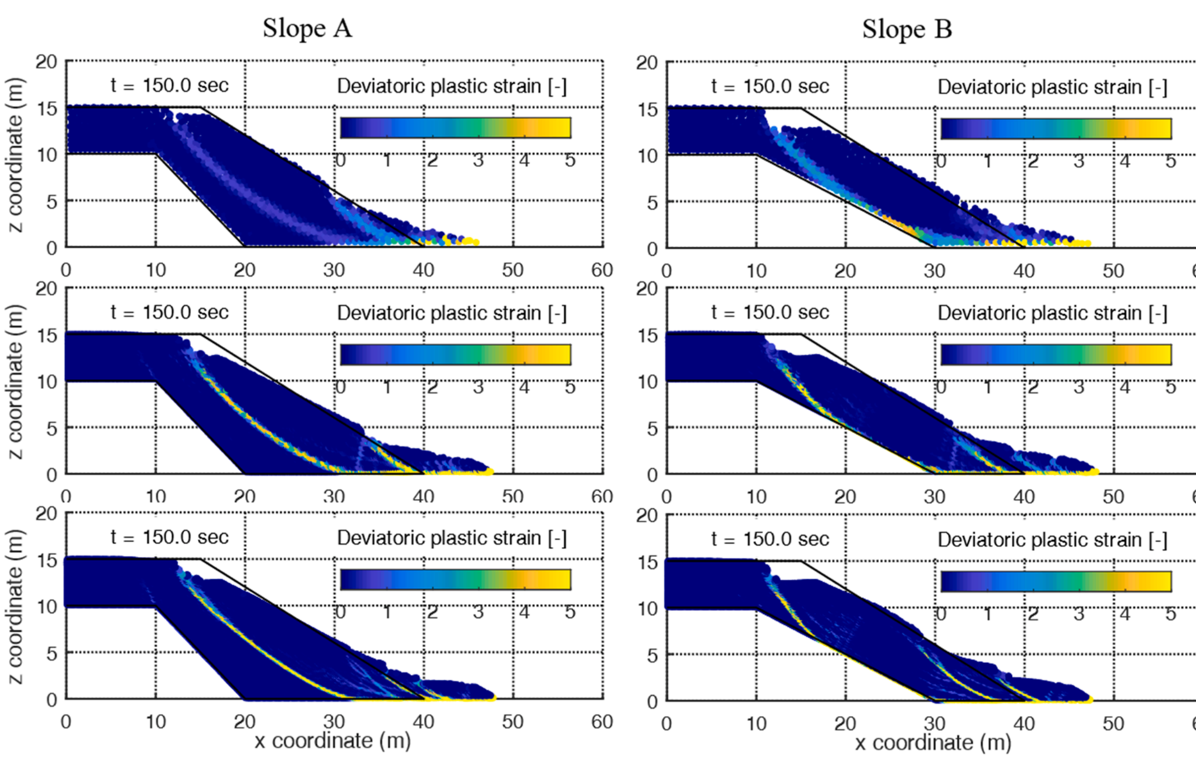

Fig. 18. Contour plot of accumulated deviatoric plastic strain with different particle resolutions: $dp = 0.5$ (first row), $dp = 0.2$ (second row), and $dp = 0.1$ (third row).

evolution and post-failure mechanisms. Due to the possible truncation of the failure surface, the geometries of bedrock could affect the shape and the movement of the slides. For the slope with deep soil cover, the local failure initiates at the toe of the slope and then propagates toward the crest; differently, for the slope with shallow soil covers, the global failure already develops at the onset of instability, indicating that the integral failure in shallow-seated slopes may give fewer warning signs than in deep-seated slopes. It should be noted that the results are also dependent of the hydrogeological characteristic of the slope and the hydro-mechanical properties of the soil. Further analyses can be investigated to provide general guidelines for the evaluation of slope stability. In addition, the proposed method is shown to be capable of capturing both the triggering of slope instability and its subsequent collapse while providing a smooth stress profile even at large deformation. Information after failure initiation can be predicted with this method and the consequence of the landslides is allowed to be evaluated.

CRediT authorship contribution statement

Ruofeng Feng: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing. **Georgios Fourtakas:** Conceptualization, Methodology, Software, Writing – review & editing. **Benedict D. Rogers:** Conceptualization, Methodology, Resources, Writing – review & editing. **Domenico Lombardi:** Conceptualization, Resources, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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